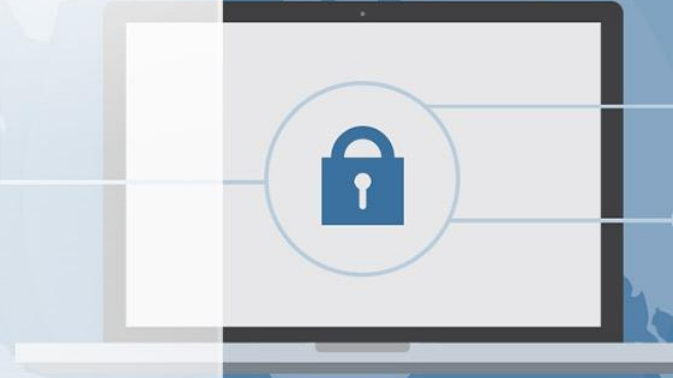




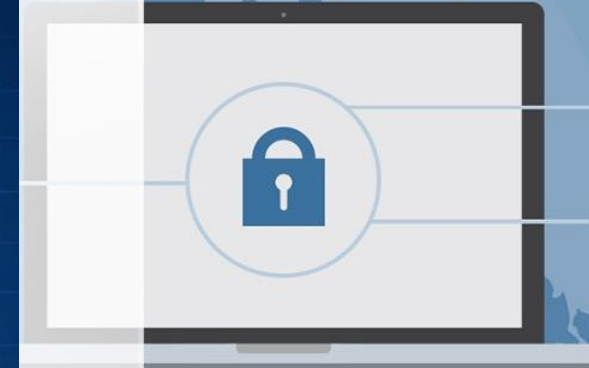
Information Security

Security Controls for CIA

- Now move from:
- **"What went wrong?"**
- to:
- **"How do we prevent it?"**



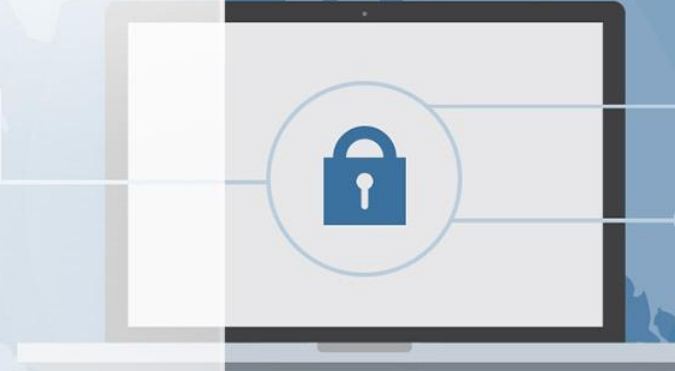
CIA Controls



Integrity Controls

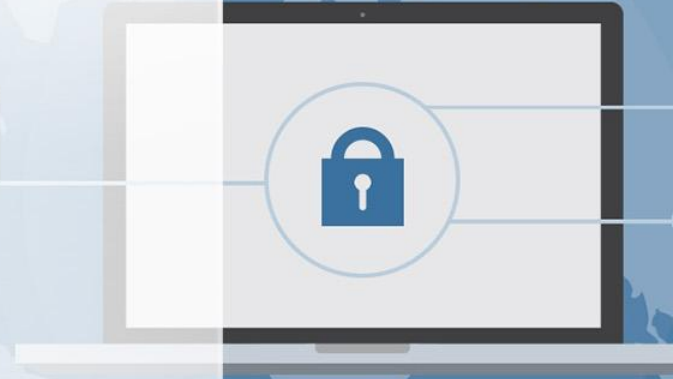
Examples:

- Hashing
- Digital signatures
- Access control
- Audit logs
- Checksums
- Version control
- Database constraints
- Change management



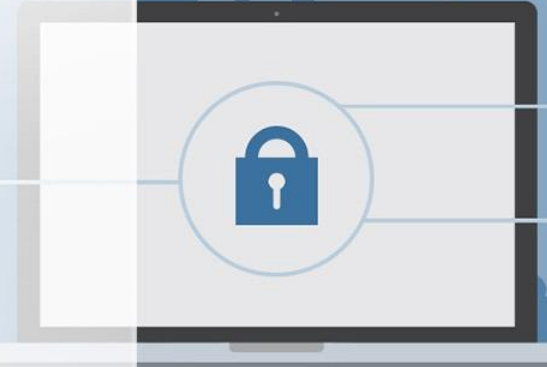
Hashing

- Converts data into a **fixed-length hash value**.
- Works like a **digital fingerprint**.
- Detect unauthorized modification or corruption.
- **Example**
 - Original File → SHA-256 → A8F5...92BC
 - After modification:
 - Modified File → SHA-256 → D72C...41AF



Digital Signatures

- A **digital signature** is a cryptographic mechanism used to verify:
- **Who sent the data**
- **Whether the data was modified**
- **Provides authenticity + integrity + non-repudiation.**
- Document → Hash → Encrypt Hash with Private Key → Digital Signature
- Receiver uses the sender's **public key** to verify it.

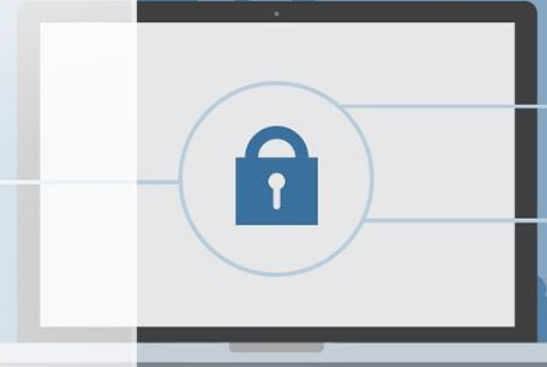


Scenario:

A student receives a digitally signed PDF certificate. After changing the student's grade, the signature becomes invalid.

Questions:

- Why did the signature become invalid?
- Which key is used to create the signature?
- Which key is used for verification?

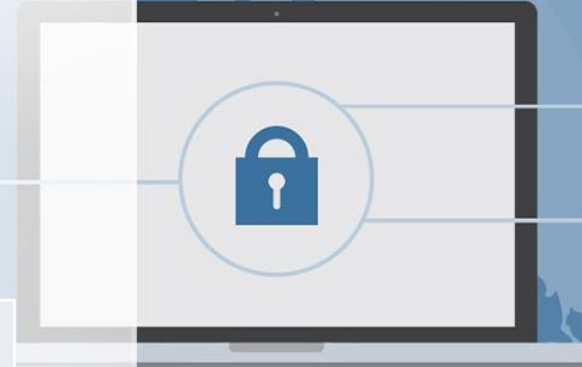


Access Control

- Access control determines who can access or modify information.
- Prevent unauthorized changes to data.

Example:

User	Access
Student	View own result
Teacher	Enter/modify marks
HOD	Approve results
IT Admin	Manage system

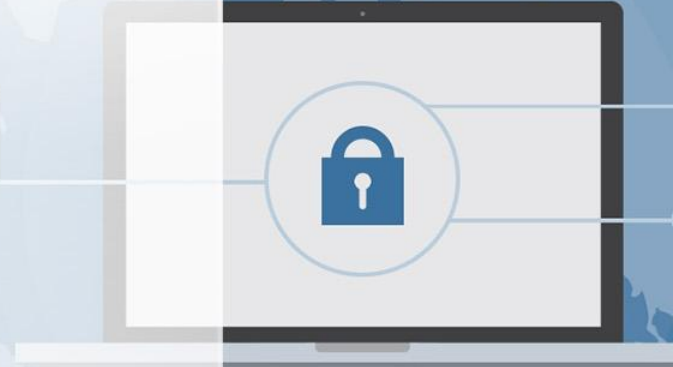


CASE STUDY

A university database allows every student to modify examination marks.

Questions:

- What security problem exists?
- Who should be allowed to modify marks?
- What access control principle should be applied?



Audit Logs

Audit logs record important activities performed on a system.

They can record:

Who performed the action

What was changed

When it happened

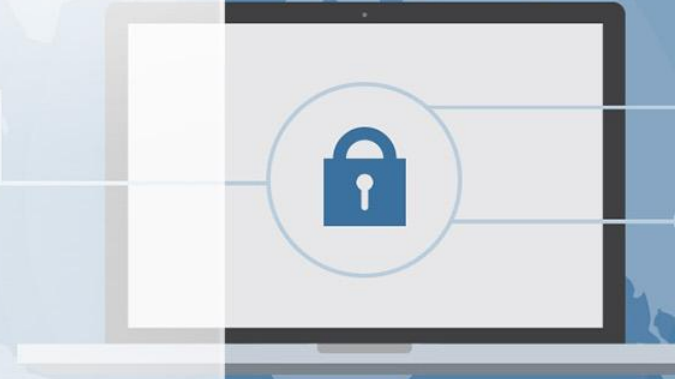
Where/from which system it happened

They Track and investigate unauthorized changes.

Example:

10:35 AM — Ali — Changed Student 105 Marks: 65 → 85

Investigators can easily identify the user responsible.



Checksums

- A **checksum** is a calculated value used to detect **errors or accidental changes** in data. **Detect data corruption during storage or transmission.**

Example

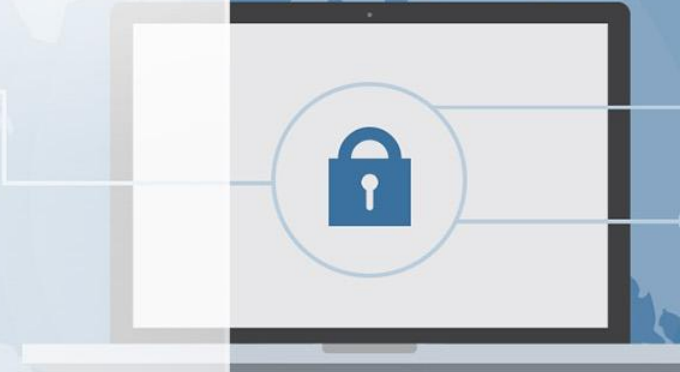
- Before transmission: File → Checksum = 45872
- After transmission: Received File → Checksum = 45872

Match = No detected change

If:

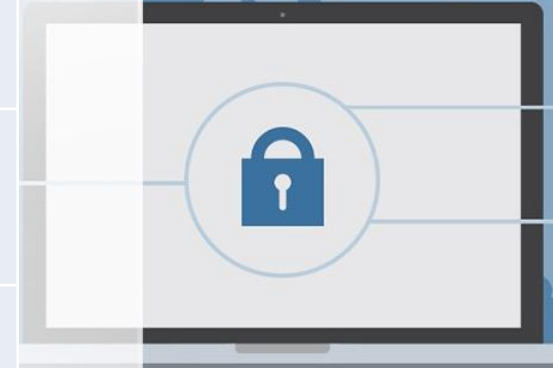
45872 $\neq$ 78234

Possible corruption/change



Hashing Vs Checksum

Hashing	Checksum
Stronger security use	Mainly error detection
Used for integrity/security	Often used for transmission/storage
SHA-256, SHA-3	CRC, simple checksums



Version Control

Version control maintains different versions of a file or system and records changes over time.

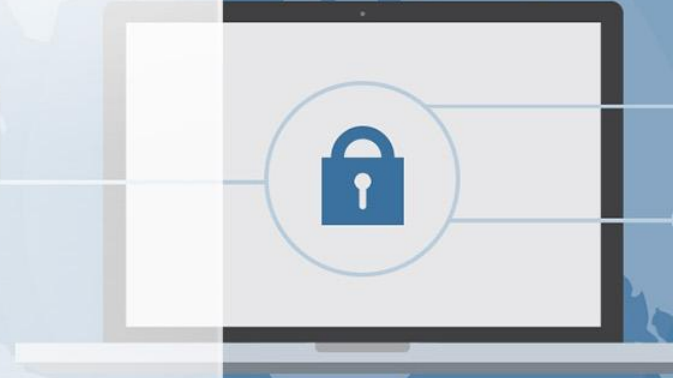
- Track changes
- Identify who changed something
- Restore previous versions
- Prevent accidental loss

Example:

Result_v1 → Result_v2 → Result_v3

If Result_v3 contains incorrect marks:

Restore **Result_v2**



Database Constraints

Database constraints are rules that prevent invalid or inconsistent data from being entered into a database.

PRIMARY KEY → Unique record identification

NOT NULL → Value cannot be empty

UNIQUE → Prevents duplicate values

FOREIGN KEY → Maintains relationships

CHECK → Ensures valid values

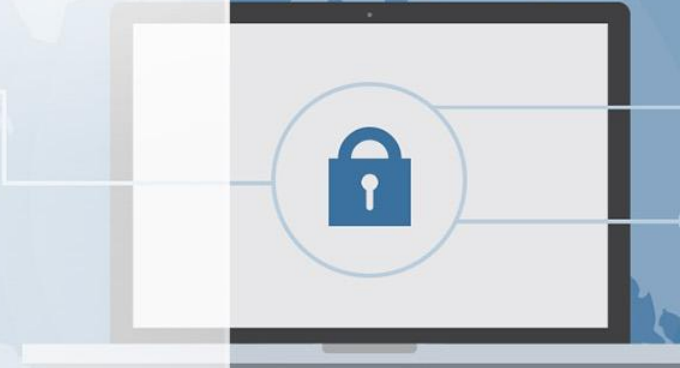
Example:

Student Marks

CHECK (Marks ≥ 0 AND Marks ≤ 100)

Marks = 85 (Right)

Marks = 150 (Wrong)



Change Management

Change management is a controlled process for making changes to systems, networks, applications, or data. **It Prevent unauthorized or poorly planned changes.**

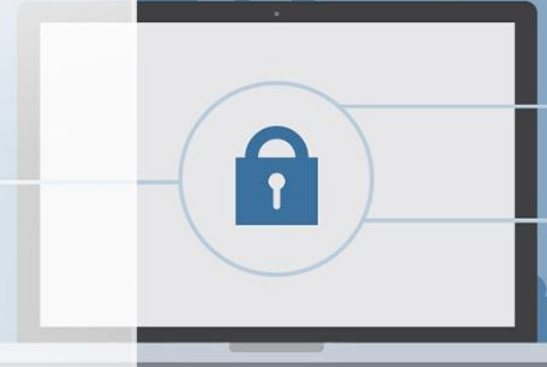
Process: Request → Review → Approval → Testing → Implementation → Documentation

Example: An administrator wants to change a university firewall rule.

✗ Directly change the rule

Instead:

Change Request → Security Review → Approval → Testing → Deployment → Log

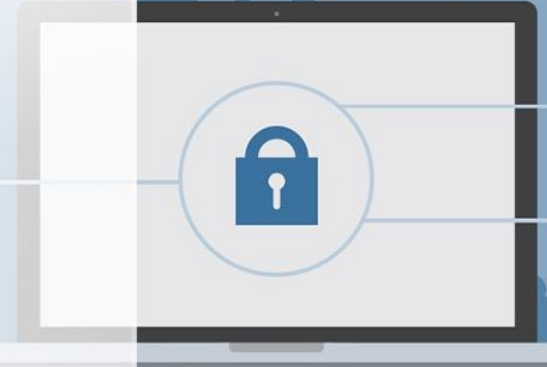


Scenario:

A network administrator changes a firewall rule without approval. The change accidentally blocks the university's online examination system.

Questions:

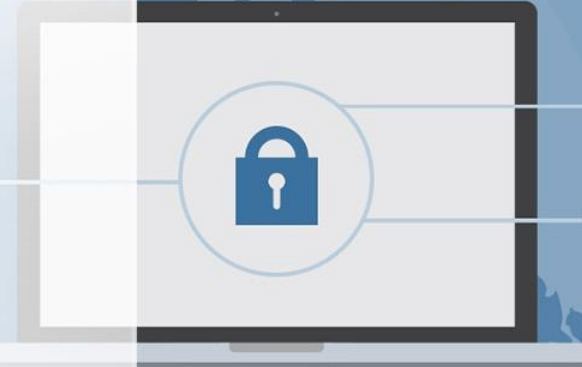
- What went wrong?
- Which change-management step was skipped?
- How could the problem have been prevented?
- Why should changes be documented?



Group Discussion

A phishing attack compromises thousand of **Google employee accounts**.

1. The attacker gains access to an internal system.
2. Customer records and important system files are modified.
3. Some changes are discovered, but the security team does not immediately know who changed what, when it happened, or whether the files are authentic.



Tasks

You are Google's Cybersecurity Expert Team.

- Identify the **integrity threats** in the scenario.
- Determine which **integrity controls** should be applied to each threat.
- Propose suitable **technical solutions**.
- Explain how you would **detect, prevent, and recover** from unauthorized changes.
- Prioritize the **three most important controls** and justify your choice.

